Rohin Joshi

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SWE Intern – Database Systems (Summer 2026)

US Citizen

EDUCATION

Carnegie Mellon University — Pittsburgh, PA, USA
Master of Science in Information Technology (GPA - 3.8) — August 2025 - May 2027
- *Relevant Coursework*: Database Systems, Distributed Systems, Computer Systems, Computer Networks

Ramaiah Institute of Technology — Bangalore, KA, India
Bachelor of Engineering in Computer Science (GPA - 3.94) — October 2020 - August 2024

EXPERIENCE

Borg Markkula Oy — Helsinki, Finland
SOFTWARE ENGINEER (PERFORMANCE) — September 2024 - October 2025
- Designed and implemented a **parallel cut-and-fill mesh processing algorithm**, using ray projections for localized mesh mutations, **reducing mesh modification runtime from 10 seconds to 30 ms per operation.**
- **Eliminated contention** in dependent operations **by decoupling transformation planning** from execution and enabling **parallel execution across worker threads**, using **lock-free queues for coordination**, reducing **runtime from 20s to 500ms**
- **Optimized CPU cache performance** on hot paths for geometric operations requiring nested loops for locality, **using stride-N patterns and tiling, increasing operation throughput by 5x.**
- Collaborated with Aalto University and StillFold to develop an origami simulator using a **multi-task U-Net (ResNet34 encoder) for feature extraction** and a **divide and conquer algorithm coupled with bidirectional EWMA for line thinning** to omit noisy artifacts and produce stable curves for hand drawn origami linework. **Published in ICCC 2026.**

Indian Institute of Science — Bangalore, India
RESEARCH ASSISTANT — January 2024 - December 2024
- Built a **generative design framework using function–behavior–structure (FBS) ontology**, using short composable chains to let the model reason from first principles and draw analogies naturally
- **Designed a two-tier agent setup**: a “generalizer” for synthesis and a “specializer” that grounds outputs using retrieval (knowledge graphs, documents, web data)
- Implemented the system as an **asynchronous, message-passing network of agents** with non-blocking execution, **cutting end-to-end latency by ~80%**
- Added a **retrieval + knowledge graph pipeline** to keep multi-step designs consistent, **reducing hallucinations by 85%.**
- **Published work in *Advanced Engineering Informatics* (Q1, IF 9.9)**

Unisys, Office of the CTO — Bangalore, India
SOFTWARE ENGINEERING INTERN — June 2023 - December 2023
- Implemented **mTLS across distributed systems and a scalable pub-sub WebSocket architecture**, **reducing latency by 75% and supporting 50K+ concurrent users** for Digital Bill of Materials (OSS under Linux Foundation)
- Built an Azure-based **asynchronous ingestion and indexing pipeline for financial data** such as balance sheets, SEC filings, etc, **enabling low-latency search** for internal systems for Unisys’ internal **financial LLM agent.**

PROJECTS

BusTub Database (C++, relational database with SQL support)
- Built a disk-oriented storage engine with slotted pages, buffer pool, and async I/O
- Designed a sharded buffer pool with ARC eviction, achieving top-5 in CMU DB leaderboard.
- Implemented a concurrent B+ tree using latch crabbing
- Engineered MVCC + lock manager for snapshot isolation with deadlock detection
- Developed a vectorized query engine with rule-based optimization ranking top 10 in CMU DB leaderboard.

GopherDB (Go, NoSQL KV-Store)
- Engineered a disk-backed NoSQL key-value store in Go, with a B+ tree index and slotted page layout for variable-length records.
- Designed a copy-on-write transaction system with atomic commit/rollback, dirty page buffering, and single-writer concurrency control.
- Built a page-oriented storage engine with freelist-based space reuse and on-disk node serialization.

SKILLS

- **Languages:** C, C++, C#, Go, Python, Java, SQL
- **Systems & Databases:** PostgreSQL, SQLite, DuckDB, Redis, MongoDB, Neo4j
- **Distributed Systems & Messaging:** Kafka, ZooKeeper, NATS, ZeroMQ
- **Cloud & Tools:** AWS, Azure, Docker, Git
- **Parallel & Systems:** OpenMP, MPI, CUDA, C# TPL